

Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this .tex file directly.

For each question (that applies), replace the “Type your response here” text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel
claims are included}
{(yes/partial/no)}
yes
```

Please make sure to:

- Replace **ONLY** the “Type your response here” text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the `\question` command or any other lines in this document.

You can `\input` this .tex file right before `\end{document}` of your main file or compile it as a stand-alone document. Check the instructions on your conference’s website to see if you will be asked to provide this checklist with your paper or separately.

1. General Paper Structure

- 1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) **partial – the differentiable engine, the gradient-fitting loss, and the identifiability diagnostic are each fully specified in equations and prose (Section 3), but not as boxed pseudocode/algorithm blocks.**
- 1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) **yes**
- 1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) **yes**

2. Theoretical Contributions

- 2.1. Does this paper make theoretical contributions? (yes/no) **no – the paper’s contributions are an empirical diagnostic (gradient-sensitivity identifiability),**

grounded in an existing theoretical quantity (Fisher information) rather than a new theoretical result, and a systematic empirical comparison of inference methods.

If yes, please address the following points:

- 2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) **NA – no theoretical contribution is claimed.**
- 2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) **NA**
- 2.4. Proofs of all novel claims are included (yes/partial/no) **NA**
- 2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) **NA**
- 2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) **yes – the Fisher-information grounding cites the relevant identifiability-analysis literature (Miao et al. 2011) even though we do not present a new theoretical result ourselves.**
- 2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) **NA**
- 2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) **NA**

3. Dataset Usage

- 3.1. Does this paper rely on one or more datasets? (yes/no) **yes**

If yes, please address the following points:

- 3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) **yes – CGMacros is the primary cohort (mixed glycemic status, free-living meals); Shanghai and Hall are chosen as external cohorts specifically because they differ from CGMacros along the two axes (population Si range, standardized vs. free-living meals) that Section 4.5’s discussion turns on.**
- 3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) **NA – no novel dataset is introduced; all three cohorts (CGMacros, ShanghaiT2DM, Hall 2018) are pre-existing published datasets.**
- 3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) **NA**
- 3.5. All datasets drawn from the existing literature (potentially including authors’ own previously published work) are accompanied by appropriate citations (yes/no/NA) **yes**

- 3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) [yes](#) – CGMacros is public on PhysioNet, ShanghaiT2DM is public on figshare, and Hall 2018's CGM traces are public PLOS Biology supplementary data.
- 3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisfying (yes/partial/no/NA) [NA](#)

4. Computational Experiments

- 4.1. Does this paper include computational experiments? (yes/no) [yes](#)

If yes, please address the following points:

- 4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) [partial](#) – final hyperparameters (Adam learning rate, optimization steps, regularization coefficient, parameter bounds) are stated in Section 3.2, but the paper does not log the full range of values explored during development before settling on them.
- 4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) [partial](#) – not embedded as a PDF appendix, but the dataset loaders (which perform all pre-processing) are included in the anonymized Code and Data Package submitted alongside this paper.
- 4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) [partial](#) – not embedded as a PDF appendix, but included as a separate anonymized Code and Data Package (simulator, all four inference methods, and all evaluation/analysis scripts) submitted alongside this paper.
- 4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) [yes](#)
- 4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) [partial](#) – the released code documents every smoothing/modeling decision inline (mirroring Section 3.1), but does not cross-reference paper section numbers directly.
- 4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results

(yes/partial/no/NA) [partial](#) – numpy/JAX/torch seeds are fixed to 42 globally (Section 3.6) and reproduce the numpy/JAX gradient and SBC-calibration results exactly. However, we identified a gap during final verification: the 5-fold cross-validation scripts assign folds using Python's built-in `hash()` on subject IDs, which is randomized per-interpreter-process (PEP 456); the codebase's seed helper sets `PYTHONHASHSEED` via `os.environ` at runtime, which does not affect the already-running process. This causes small (<1 minute MAE, a few percentage points on paired win rates) run-to-run variation in the k-fold tables specifically, which we verified does not change any qualitative conclusion (re-running produced the same paired win rate against grid search reported in the paper). We report this honestly as a reproducibility gap rather than overstating full determinism; fixing it (e.g. a stable/non-randomized hash of the subject ID) is a one-line change we have not yet made to the released code.

- 4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) [partial](#) – software stack and CPU-only requirement are stated (Section 3.6); exact CPU model, memory, and OS are not specified as none of the experiments in this paper are compute- or memory-bound at this problem size (per-subject fits, 45–97 subjects).
- 4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) [yes](#)
- 4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) [yes](#) – 5-fold cross-validation at a single fixed seed, with 1000-resample bootstrap for correlation confidence intervals (Section 3.6).
- 4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) [yes](#) – bootstrap confidence intervals, fold-level standard deviations, and paired per-subject win rates are reported throughout Section 4.
- 4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) [partial](#) – correlation coefficients carry bootstrap confidence intervals, and prediction comparisons are reported as paired per-subject win rates, but no formal paired significance test (e.g., Wilcoxon signed-rank) is applied to the prediction comparisons themselves.
- 4.13. This paper lists all final (hyper-)parameters used

for each model/algorithm in the paper's experiments
(yes/partial/no/NA) [yes](#)